

This lab session covers Django Models (Part 3). By the end of this session, students should have a very good understanding of how to make advanced queries using annotations.

Pre-lab Preparation:

1. The following DB models scheme

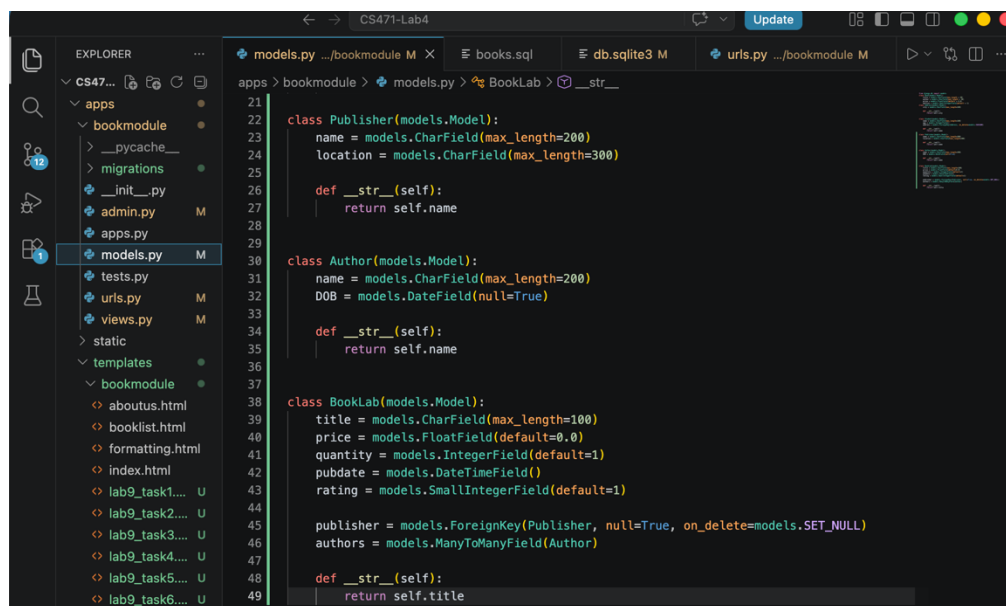
```
class Publisher(models.Model):
    name = models.CharField(max_length=200)
    location = models.CharField(max_length=300)

class Author(models.Model):
    name = models.CharField(max_length=200)
    DOB = models.DateField(null=True)

class Book(models.Model):
    title = models.CharField(max_length= 100)
    price = models.FloatField(default=0.0)
    quantity = models.IntegerField(default=1)
    pubdate = models.DateTimeField()
    rating = models.SmallIntegerField(default = 1)

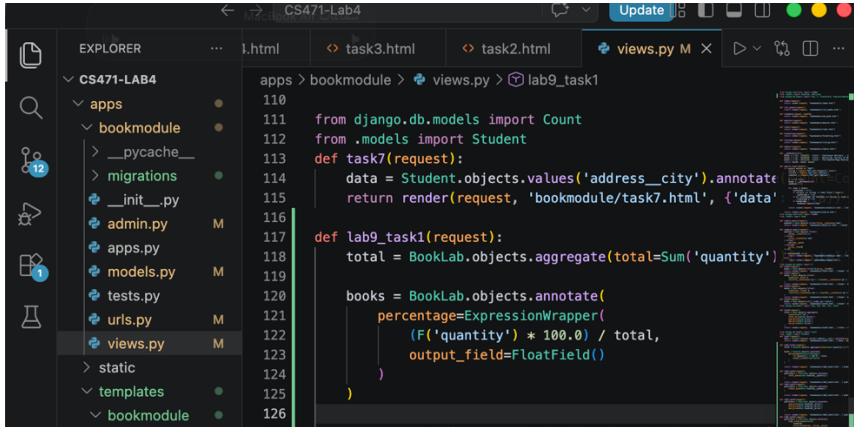
    publisher = models.ForeignKey(Publisher, null=True, on_delete=models.SET_NULL)
    authors = models.ManyToManyField(Author)
```

2. Use fake data from mockaroo.com website or any other methods to fill tables above. Make sure that you handle the Book.publisher and Book.authors fields correctly.

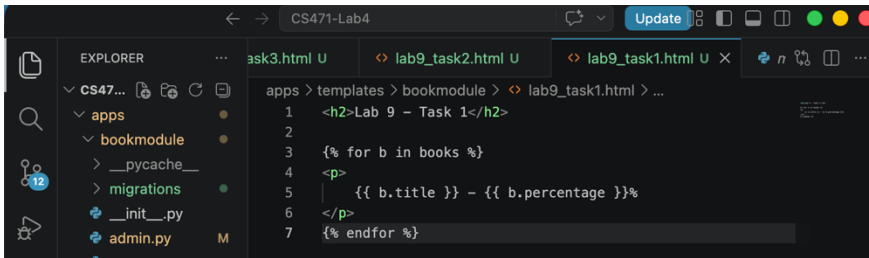


Lab Activities:

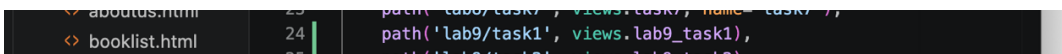
Task 1: Create a URL `‘/books/lab9/task1’` with any necessary HTML file and view function to list books along with percentage availability. For example, if the number of books is 350 and the quantity of a book is 7, then the percentage of availability is 2%. Use transient field for the percentage.



```
apps > bookmodule > views.py > lab9_task1
110
111 from django.db.models import Count
112 from .models import Student
113 def task7(request):
114     data = Student.objects.values('address__city').annotate(
115         return render(request, 'bookmodule/task7.html', {'data':
116
117
118 def lab9_task1(request):
119     total = BookLab.objects.aggregate(total=Sum('quantity'))
120
121     books = BookLab.objects.annotate(
122         percentage=ExpressionWrapper(
123             (F('quantity') * 100.0) / total,
124             output_field=FloatField()
125     )
126
```



```
apps > templates > bookmodule > lab9_task1.html > ...
1 <h2>Lab 9 – Task 1</h2>
2
3 {% for b in books %}
4 <p>
5     {{ b.title }} – {{ b.percentage }}%
6 </p>
7 {% endfor %}
```



```
24 path('lab9/task1', views.lab9_task1),
25
```



Lab 9 - Task 1

Book 1 - 71.42857142857143%

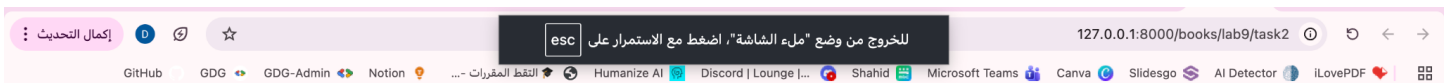
Book 2 - 28.571428571428573%

Task 2: Create a URL `‘/books/lab9/task2’`, HTML file, and query list books that all publishers, annotated with their respective total book stock.

```
def lab9_task2(request):
    publishers = Publisher.objects.annotate(
        total_books=Sum('booklab__quantity')
    )
    return render(request, 'bookmodule/lab9_task2.html', {'publishers': publishers})
```

```
lab9_task2.html U
1 <h2>Lab 9 – Task 2</h2>
2
3 {% for p in publishers %}
4 <p>
5     {{ p.name }} – {{ p.total_books }}
6 </p>
7 {% endfor %}
```

```
path('lab9/task2', views.lab9_task2),
```



Lab 9 - Task 2

Pearson - 5

O'Reilly - None

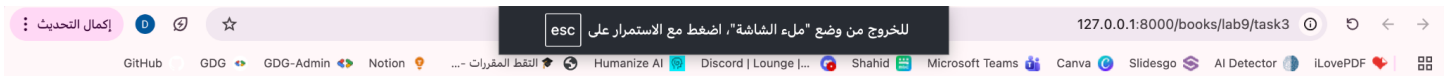
McGraw Hill - 2

Task 3: (opposite to Task 2) Create a URL `‘/books/lab9/task3’`, HTML file, and a query to get oldest book of any of those managed by every publisher.

```
def lab9_task3(request):
    publishers = Publisher.objects.annotate(
        oldest_book=Min('booklab__pubdate')
    )
    return render(request, 'bookmodule/lab9_task3.html', {'publishers': publishers})
```

```
lab9_task3.html U
1 <h2>Lab 9 – Task 3</h2>
2
3 {% for p in publishers %}
4 <p>
5     {{ p.name }} – {{ p.oldest_book }}
6 </p>
7 {% endfor %}
```

```
path('lab9/task3', views.lab9_task3),
```



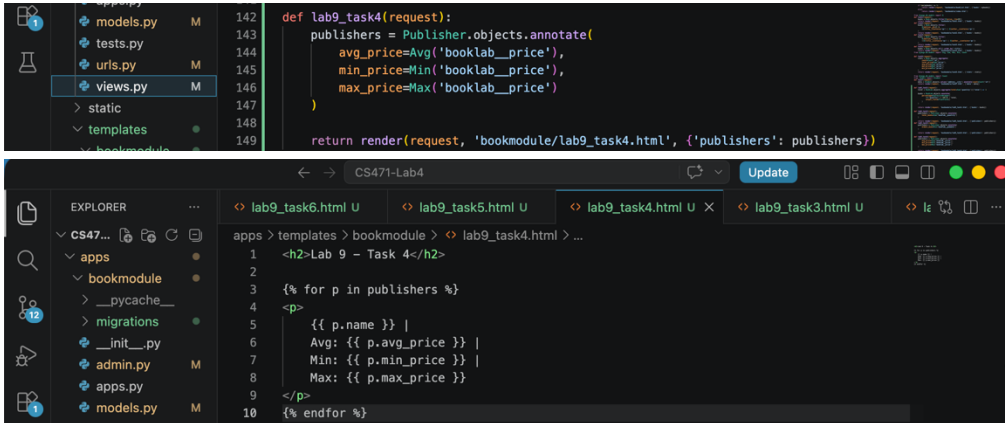
Lab 9 - Task 3

Pearson - April 15, 2026, 6 a.m.

O'Reilly - None

McGraw Hill - April 21, 2026, noon

Task 4: Create a URL `/books/lab9/task4`, HTML file, and query to calculate the average, min, and max price of books for each publisher.



```
def lab9_task4(request):
    publishers = Publisher.objects.annotate(
        avg_price=Avg('booklab__price'),
        min_price=Min('booklab__price'),
        max_price=Max('booklab__price')
    )
    return render(request, 'bookmodule/lab9_task4.html', {'publishers': publishers})
```

```
<h2>Lab 9 - Task 4</h2>
{% for p in publishers %}
<p>
    {{ p.name }} |
    Avg: {{ p.avg_price }} |
    Min: {{ p.min_price }} |
    Max: {{ p.max_price }}
</p>
{% endfor %}
```

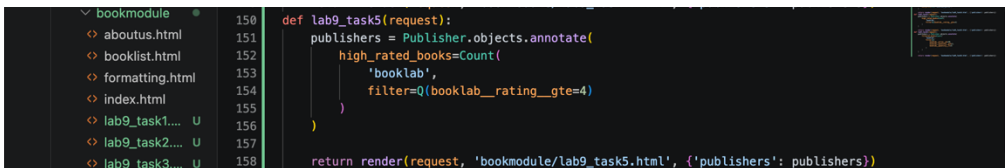
```
path('lab9/task4', views.lab9_task4),
```



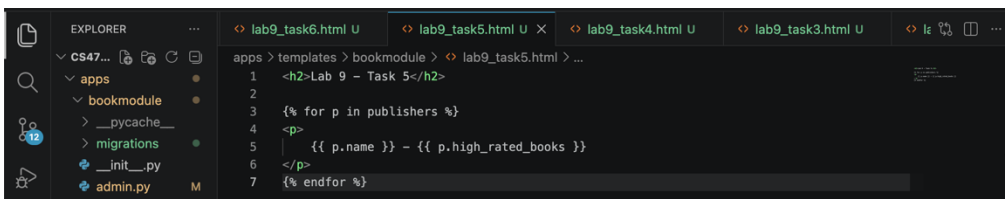
Lab 9 - Task 4

Pearson | Avg: 60.0 | Min: 60.0 | Max: 60.0
O'Reilly | Avg: None | Min: None | Max: None
McGraw Hill | Avg: 100.0 | Min: 100.0 | Max: 100.0

Task 5: Create a URL `/books/lab9/task5`, HTML file, and a query to list publishers with a count of highly rated books, with their names and quantity.

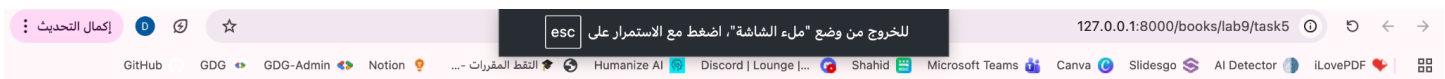


```
def lab9_task5(request):
    publishers = Publisher.objects.annotate(
        high_rated_books=Count(
            'booklab',
            filter=Q(booklab__rating__gte=4)
        )
    )
    return render(request, 'bookmodule/lab9_task5.html', {'publishers': publishers})
```



```
<h2>Lab 9 - Task 5</h2>
{% for p in publishers %}
<p>
    {{ p.name }} - {{ p.high_rated_books }}
</p>
{% endfor %}
```

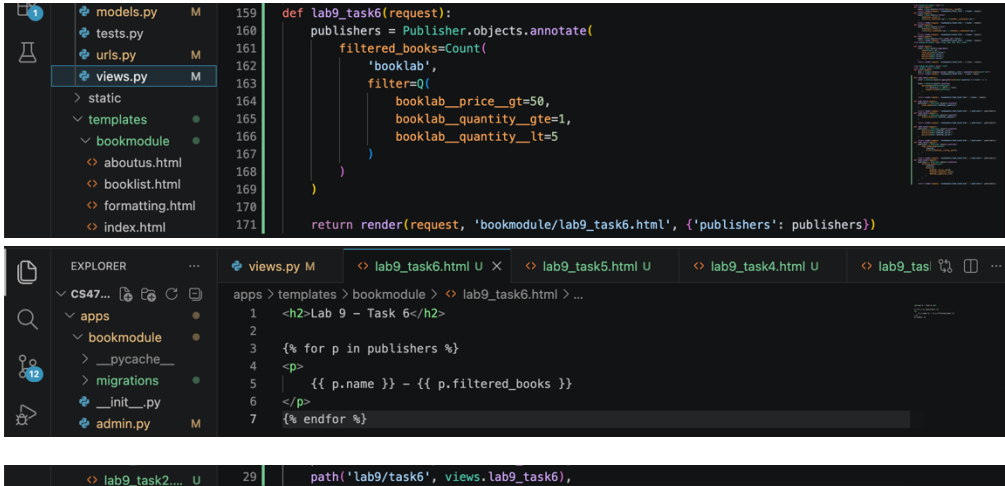
```
path('lab9/task5', views.lab9_task5),
```



Lab 9 - Task 5

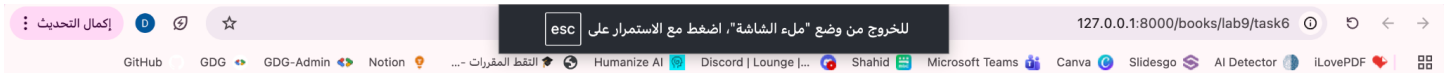
Pearson - 1
O'Reilly - 0
McGraw Hill - 1

Task 6: Create a URL `‘/books/lab9/task6’`, HTML file, and a query to count of books for each publisher, filter only books with price greater than 50 and quantity less than 5 and greater than or equal to 1.



```
def lab9_task6(request):
    publishers = Publisher.objects.annotate(
        filtered_books=Count(
            'booklab',
            filter=Q(
                booklab__price__gt=50,
                booklab__quantity__gte=1,
                booklab__quantity__lt=5
            )
        )
    )
    return render(request, 'bookmodule/lab9_task6.html', {'publishers': publishers})
```

```
<h2>Lab 9 - Task 6</h2>
{% for p in publishers %}
<p>
    {{ p.name }} - {{ p.filtered_books }}
</p>
{% endfor %}
```



Lab 9 - Task 6

Pearson - 0

O'Reilly - 0

McGraw Hill - 1